

ECE 228 | Group 5

Presentation

ALPR for Egyptian Plates



Group 5

INTRODUCTION: WHAT IS ALPR?

Understanding ALPR

Automatic License Plate Recognition uses OCR on images to read vehicle registration plates. It is vital for traffic monitoring and automated toll systems.

The Egyptian Challenge

Egyptian plates feature complex bilingual layouts including Arabic letters and Hindi (Eastern Arabic) numerals, requiring robust interpretability.

DATASET DESCRIPTION

Sample Count: 50 Distinct Images

Diverse real-world scenarios to test robustness:



Far and close proximity plates



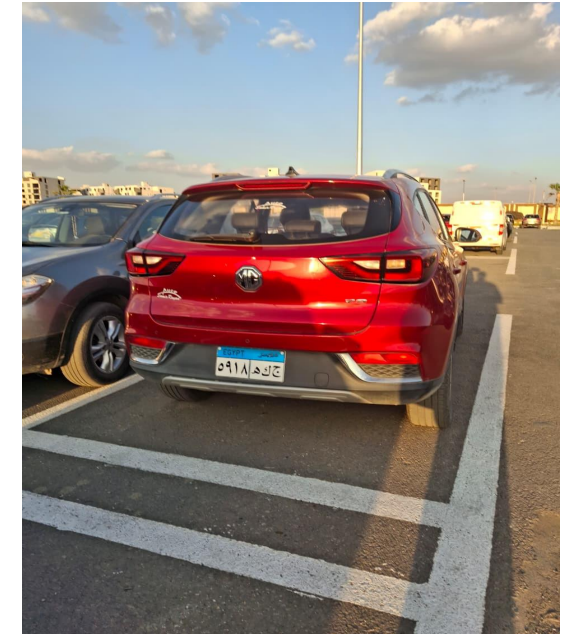
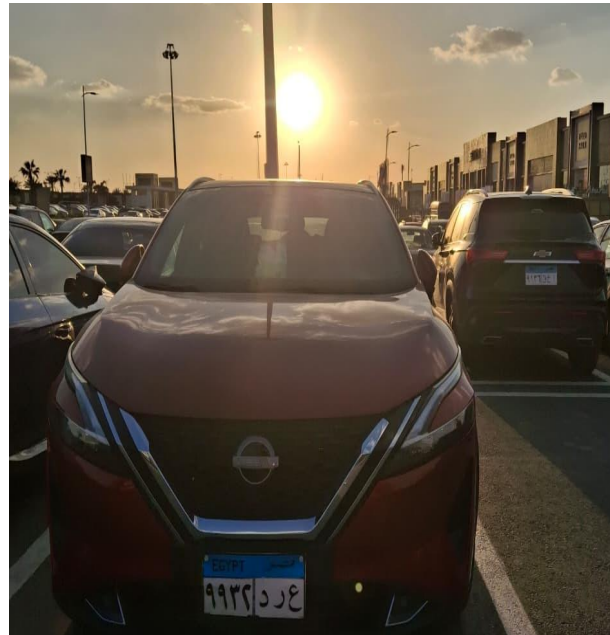
Faded paint and character wear



Varying camera angles (Skewness)



Lighting variations and noise



COMPUTER VISION PIPELINE



1. Preprocessing

Detection,
Cropping,
enhancing



2. Templates

Generating
templates for both
digits and letters



3. Segmentation

Connected
components and
Fragment
reconstruction,
using binarization
method



4. Matching and Score

Template
matching for
recognition to
detect the letters
or digits

STEP 1: PLATE DETECTION

-  **Smart Cropping:** Plate is always in some specific area in all the images, so we crop the photo in that area
-  **Edge detection:** Detects the 4 edges of the plate to cut it perfectly using a lot of conditions
-  **Enhancing:** Removes edge noise that could trigger false positive segments
-  **Divider Killer:** Targets and removes horizontal structural lines specific to Egyptian plates



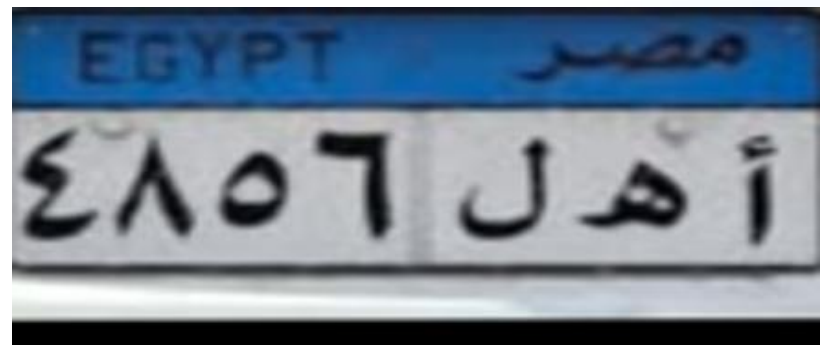
STEP 2: NOISE & MORPHOLOGY

Noise, Faded, Low Quality

Plates are not always perfect, some of them doesn't have clear letters, so are faded, some are low quality due to far away cars.

Correcting Angles

Some plates are captured with angles in it, that are not directly horizontal to the screen, so we made a code that enhances the images and fix all these problems



RESULTS SUMMARY :



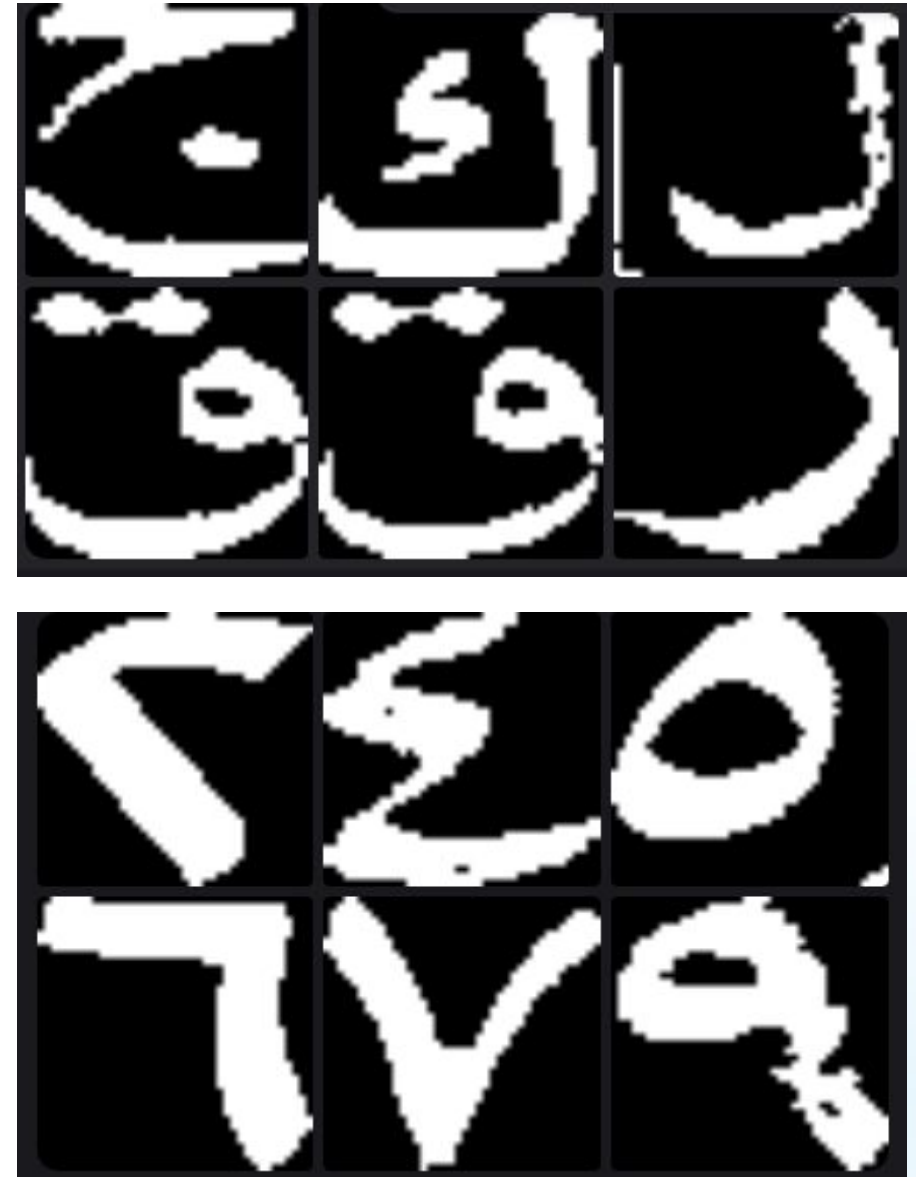
STEP 3: Templates

Why?: It's a smart way to detect the letters and digits using correlation only and without using an AI or models

How?: We got some test plates to generate the templates from it and used totally another data set for the testing, each template is 40*40 pixel

Logic: Each letter and digit is stored in an array with its name, when doing the matching, it finds the best score with the template and takes it as the letter or digit.

How Many?: For each letter we made about 4-5 templates in different situations and shapes so all possible cases can be detected



STEP 4: BINARIZATION

Processing Logic:

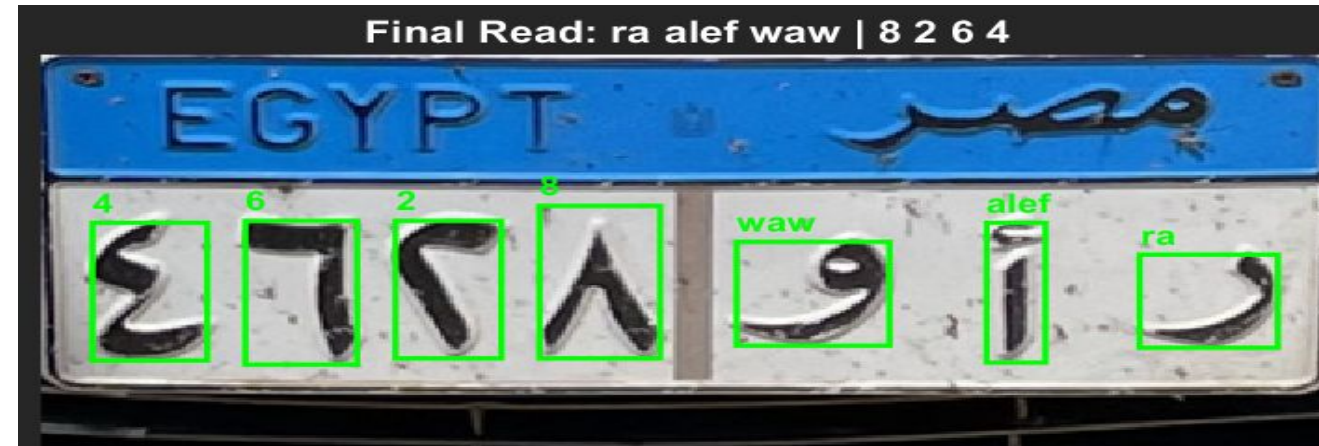
- ← We use binarization so separate the letters and digits from each others because of their unique white color in the binarization.
- ❏ Also there is a part in the code to remove the white vertical line between the letters and digits, to not detect it as a separate letters



STEP 5: TEMPLATE MATCHING

Processing Logic:

- ← Comparing the binarization and templates to get the letters and digits, choosing the best scoring result
- ☐ Making green boxes about each segmented letter or digit, then doing the matching, then writing the result above the image and write it on the screen



RESULTS & ACCURACY



**Full plate exact match refers to success rate on a specifically cleaned subset.*

DISADVANTAGES

Problem 1:

Not clear letters causes errors in the segmentation and detection because the letters are not clear and faded

Problem 2:

Angled plates, this made a lot of problems in the detection because the digits and letters are not zero angled so it's hard to be detected like the correct angled letters and



WORK DISTRIBUTION

Team Member	Responsibility	Key Deliverable
Mostafa Alaa Mohamed	Lead Developer	Plate Detection Code
Abdallah Ahmed Mohamed	Image Processing	Preprocessing & Thresholding
Mohamed Arafa Elsayed	Data Collector	Dataset & Photo Taking
Khaled Osama Abdelsalam	Validation	Accuracy & Photo Taking
Omar Mohamed Abdelaziz	Debugger	Template Creation, Error Analysis
Arwa Bahaa Hussein	Researcher	Report, Organizing Files
Jasmine Ahmed Abdelaal	Researcher	Presentation, Template Creation

Q&A

THANK YOU :)